

# PROJECT REPORT

## Cover Page

**Project Title:** PharmaLLM: A Lightweight Domain-Tuned Medical Assistant

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**Course/Program:** ITCS 5154 - Applied Machine Learning

**Year:** 2026

## Primary Paper Information

**Authors:** Ayesha Azam, Zubaira Naz, Muhammad Usman Ghani Khan

**Title:** PharmaLLM: A Medicine Prescriber Chatbot Exploiting Open-Source Large Language Models

**Year:** 2024

**Venue:** Human-Centric Intelligent Systems

**Link:** <https://link.springer.com/article/10.1007/s44230-024-00085-z>

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## Introduction

### Problem Statement

General language models can answer health questions, but they often miss medication-specific details or respond with wording that is too broad for focused drug queries. Our project targets that gap. We built a medicine-centered assistant that answers questions about uses, composition, and side effects with tighter output control, following the PharmaLLM reproduction goal from Azam et al. (2024).

### Motivation

Medication guidance is a hard space because users want quick, readable answers while the domain itself is technical. At the same time, many student teams do not have access to large GPU servers. So we focused on a practical question: can a small open model, tuned carefully, become useful for medicine Q and A without expensive infrastructure, as explored in the PharmaLLM direction (Azam et al., 2024)?

### Open Questions in the Domain

How much reliability can parameter-efficient fine-tuning add before full retraining becomes necessary? Can compact models stay stable when prompts are vague or mixed-language? How do we balance short responses with enough safety context? And when we add speech and translation modules, how much error do those modules introduce into the final answer?

## Brief Overview of Approach

This project is a reproduction and implementation study based on Azam, Naz, and Khan (2024). We followed the core PharmaLLM direction using TinyLlama-1.1B with LoRA fine-tuning, then built an end-to-end working pipeline that includes dataset preparation, training, evaluation, and deployment through Flask and React. We also integrated speech-to-text, translation, and text-to-speech features to match the multimodal goal described in the proposal.

This system is a research prototype for educational use. It is not intended for clinical deployment, and responses are framed with safety language that advises users to consult a qualified clinician.

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## Backgrounds/Related Work

### Related Work 1: LoRA for Efficient Adaptation

**Citation:** Hu et al., 2021, *LoRA: Low-Rank Adaptation of Large Language Models* (arXiv)

LoRA introduces trainable low-rank matrices while freezing most base model parameters. In my interpretation, the key contribution is that high-impact adaptation can be achieved with only a small fraction of trainable weights.

#### Pros

Strong parameter efficiency, lower memory demand, and simple adapter-based deployment.

#### Cons

Performance can drop when adapter rank and target modules are not tuned well, and in some tasks full fine-tuning can still win.

#### Relation to Our Method

Our pipeline directly follows this idea: TinyLlama remains mostly frozen while LoRA layers encode medicine-domain behavior.

### Related Work 2: QLoRA for Low-Memory Fine-Tuning

**Citation:** Dettmers et al., 2023, *QLoRA: Efficient Finetuning of Quantized LLMs* (arXiv)

QLoRA shows that 4-bit quantization plus adapter training can preserve useful downstream performance while reducing memory pressure. My takeaway is that quantization and parameter-efficient adaptation are complementary rather than competing strategies.

#### Pros

Makes large-model adaptation possible on consumer hardware while preserving practical model quality.

#### Cons

Adds engineering complexity and introduces quantization hyperparameters that can affect convergence.

**Relation to Our Method**

Our training setup uses 4-bit NF4 quantization and LoRA together to run in constrained hardware environments.

**Related Work 3: Compact Open Models for Instruction Tasks**

**Citation:** Zhang et al., 2024, *TinyLlama: An Open-Source Small Language Model* (GitHub Project)  
TinyLlama demonstrates that smaller open models can still provide useful instruction-following behavior, especially when adapted to a specific domain.

**Pros**

Lightweight enough for student projects and easier to deploy in resource-limited environments.

**Cons**

More fragile under challenging prompts and usually weaker on broad reasoning than larger models.

**Relation to Our Method**

TinyLlama is our base model, chosen to prioritize efficient experimentation and practical deployment.

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**Methods**

**Overall Framework**

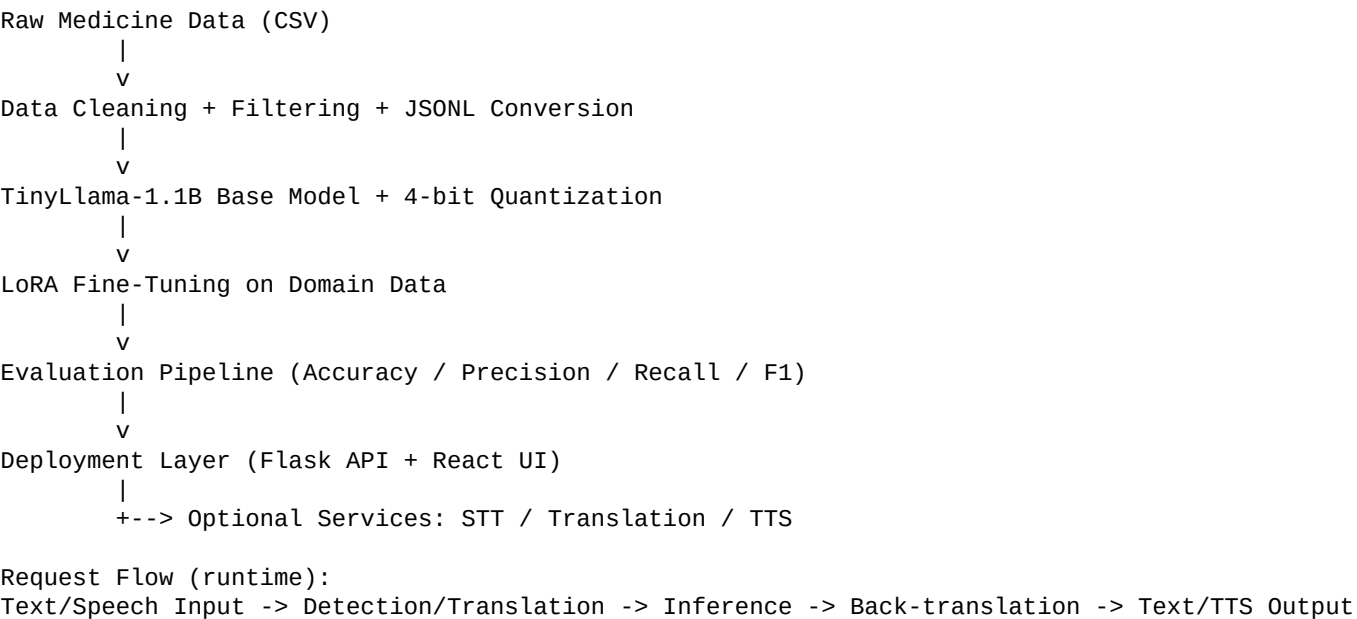
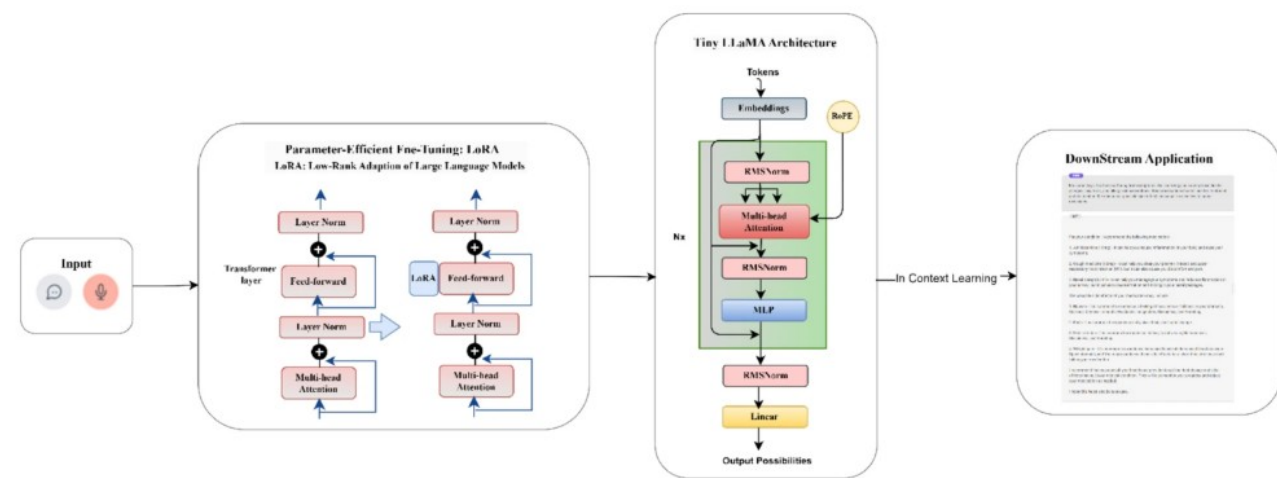


Figure 1 presents the end-to-end architecture view used in this report.

**Fig. 1**  
**From: PharmaLLM: A Medicine Prescriber Chatbot Exploiting Open-Source Large Language Models**



Architecture Diagram of Proposed PharmaLLM

Figure: Architecture diagram of proposed PharmaLLM

**Algorithm and Training Details**

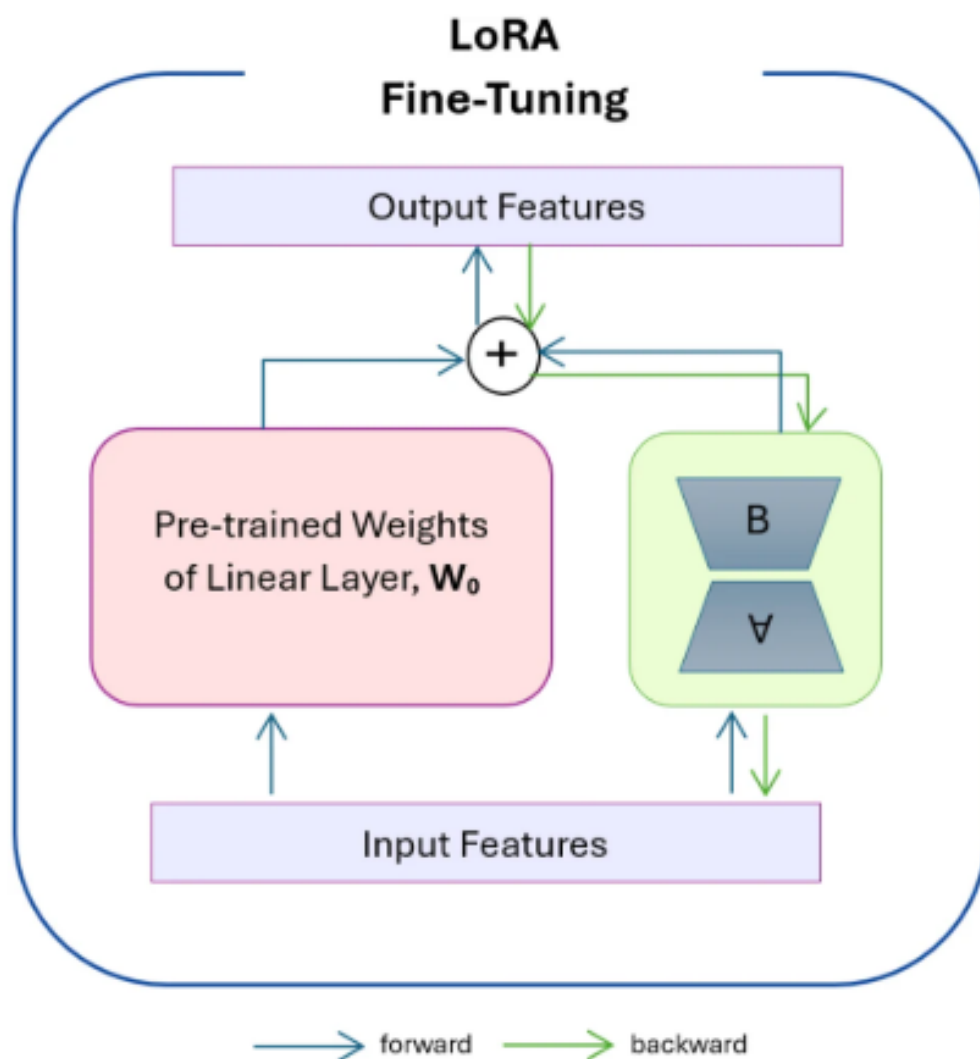
The base model is TinyLlama-1.1B-Chat (Zhang et al., 2024). We apply LoRA adapters (Hu et al., 2021) on top of a 4-bit quantized backbone using NF4 plus double quantization in the QLoRA style (Dettmers et al., 2023). In our configuration, LoRA uses rank  $r=8$  and  $\text{lora\_}\alpha=16$ , with gradient accumulation configured to reach an effective batch size of 16. Training uses a max sequence length of 256 to stay within memory limits on low-VRAM hardware.

At inference time, the backend loads the base model with LoRA adapters and runs generation in fp16/fp32 as supported by device settings, rather than serving from a quantized training graph. During generation, we keep temperature low, apply repetition control, and use deterministic decoding to reduce unstable outputs.

Figure 2 illustrates the LoRA adaptation concept used in the training pipeline.

**Fig. 2**

**From: PharmaLLM: A Medicine Prescriber Chatbot Exploiting Open-Source Large Language Models**



LoRA Fine-tuning

Figure: LoRA fine-tuning concept diagram

Figure 3 shows the algorithm-style training flow used for TinyLlama fine-tuning in the reproduced pipeline.

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**Algorithm 1 Tiny Llama Fine Tuning**

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**Input:** user query (text, speech)

**Output:** Diagnosis or treatment suggestion

**Start**

```
1:  $lr \leftarrow 2 \times e^{-4}$  ▷  $lr$  is Initial_Learning_rate
2:  $Batch\_Size \leftarrow 12$ 
3:  $Number\_of\_Epochs \leftarrow 3$ 
4:  $Trainer \leftarrow sft$  ▷  $sft$  is supervised fine-tuning
5:  $peft \leftarrow \text{True}$  ▷ PEFT is sparse parameter-efficient fine-tuning
6:  $model \leftarrow \text{loadPretrainedModel}(\text{"Tiny LLaMA"})$ 
7:  $optimizer \leftarrow \text{initializeOptimizer}(model.parameters(), learningRate)$ 
8:  $trainDataset \leftarrow \text{loadDataset}(\text{"path/to/train/data"})$ 
9: if  $peft$  is True
10:   then  $model \leftarrow \text{applyPEFT}(model)$ 
11: end if
12: for  $epoch \leftarrow 1$  to  $epochs$  do
13:    $\text{shuffle}(trainDataset)$ 
14:   for  $batch$  in  $\text{createBatches}(trainDataset, batchSize)$  do
15:      $outputs \leftarrow model.forward(batch["input"])$ 
16:      $loss \leftarrow \text{computeLoss}(outputs, batch["target"])$ 
17:      $loss.backward()$ 
18:     if  $trainer$  is " $sft$ " then
19:        $\text{applySFT}(model)$ 
20:     end if
21:      $optimizer.step()$ 
22:      $optimizer.zeroGrad()$ 
23:   end for
24:    $evalLoss \leftarrow \text{evaluateModel}(model, evalDataset)$ 
25:    $\text{print}(\text{"Epoch", epoch, "Validation Loss:"}, evalLoss)$ 
26:    $\text{saveCheckpoint}(model, \text{"checkpoint\_epoch\_"} + epoch)$ 
27: end for
28:  $\text{saveModel}(model, \text{"final\_tiny\_llama\_model"})$ 
```

**End**

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Figure: Algorithm for TinyLlama fine-tuning

## Data Preprocessing

We start from a medicine CSV and validate required columns such as medicine name, composition, uses, and side effects. We compute a satisfaction proxy from review percentages, filter out low-score rows, and convert each remaining row into a consistent natural-language training example. After that, we shuffle and split into train and validation JSONL files. This keeps the format stable so the model learns a repeatable answer pattern rather than noisy free-form behavior.

Consistent with the project proposal and slide deck, we use a satisfaction filter threshold of 70% or higher to prioritize cleaner records for fine-tuning.

Figure 4 provides a snapshot of exploratory data analysis on the medicine table used before preprocessing and filtering.

Fig. 3

From: PharmaLLM: A Medicine Prescriber Chatbot Exploiting Open-Source Large Language Models

| Medicine Name                      | Composition                                                                  | Uses                                                                                     | Side effects                                                                | Image URL                                                                   | Manufacturer                      | Excellent | Average | Poor | Review % |
|------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------|-----------|---------|------|----------|
| 1 Avastin 400mg Injection          | Bevacizumab (400mg)                                                          | Cancer of colon and rectum Non-small cell lung                                           | Rectal bleeding Taste change Headache No                                    | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Roche Products India Pvt Ltd      | 22        | 56      | 22   |          |
| 2 Augmentin 625 Duo Tablet         | Amoxycillin (500mg) + Clavulanic Acid (125mg)                                | Treatment of Bacterial infections                                                        | Vomiting Nausea Diarrhea Mucocutaneous                                      | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Glaxo SmithKline Pharmaceuticals  | 47        | 35      | 18   |          |
| 4 Azithral 500 Tablet              | Azithromycin (500mg)                                                         | Treatment of Bacterial infections                                                        | Nausea Abdominal pain Diarrhea                                              | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Alembic Pharmaceuticals Ltd       | 39        | 40      | 21   |          |
| 5 Ascoril LS Syrup                 | Ambroxol (30mg/5ml) + Levosalbutamol (1mg/5ml) + Guafenesin (50mg/5ml)       | Treatment of Cough with mucus                                                            | Nausea Vomiting Diarrhea Upset stomach                                      | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Glenmark Pharmaceuticals Ltd      | 24        | 41      | 35   |          |
| 6 Aclorac 150 Tablet               | Ranitidine (150mg)                                                           | Treatment of Gastroesophageal reflux disease (Headache Diarrhea Gastrointestinal disturb | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Cadila Pharmaceuticals Ltd                                                  | 34                                | 37        | 29      |      |          |
| 7 Allegra 120mg Tablet             | Fexofenadine (120mg)                                                         | Treatment of Sneezing and runny nose due to al                                           | Headache Drowsiness Dizziness Nausea                                        | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Sanofi India Ltd                  | 35        | 42      | 23   |          |
| 8 Avil 25 Tablet                   | Pheniramine (25mg)                                                           | Treatment of Allergic conditions Treatment of R                                          | Sedation                                                                    | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Sanofi India Ltd                  | 40        | 34      | 26   |          |
| 9 Aricep 5 Tablet                  | Donepezil (5mg)                                                              | Alzheimer's disease                                                                      | Common cold Urinary incontinence Rash N                                     | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Eisai Pharmaceuticals India Pvt L | 43        | 28      | 29   |          |
| 10 Amoxyclev 625 Tablet            | Amoxycillin (500mg) + Clavulanic Acid (125mg)                                | Treatment of Bacterial infections                                                        | Vomiting Nausea Diarrhea Mucocutaneous                                      | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Abbott                            | 36        | 43      | 21   |          |
| 11 Atarax 25mg Tablet              | Hydroxyzine (25mg)                                                           | Treatment of AnxietyTreatment of Skin conditio                                           | Sedation Nausea Vomiting Upset stomach                                      | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Dr Reddy's Laboratories Ltd       | 35        | 41      | 24   |          |
| 12 Azee 500 Tablet                 | Azithromycin (500mg)                                                         | Treatment of Bacterial infections                                                        | Nausea Abdominal pain Diarrhea                                              | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Cipla Ltd                         | 37        | 38      | 25   |          |
| 13 Anovate Cream                   | Phenylephrine (0.10% w/w) + Beclomethasone (0.025% w/w) + Lidocaine (2.50% v | Treatment of Piles                                                                       | Application site reactions burning irritation                               | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | USV Ltd                           | 26        | 43      | 31   |          |
| 14 Allegra H Tablet                | Montelukast (10mg) + Fexofenadine (120mg)                                    | Treatment of Sneezing and runny nose due to al                                           | Nausea Diarrhea Vomiting Skin rash Flu like                                 | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Sanofi India Ltd                  | 38        | 44      | 18   |          |
| 15 Ascoril D Plus Syrup Sugar Free | Phenylephrine (5mg) + Chlorpheniramine Maleate (2mg) + Dextromethorphan H    | Treatment of Dry cough                                                                   | Nausea Vomiting Loss of appetite Headache                                   | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Glenmark Pharmaceuticals Ltd      | 31        | 44      | 25   |          |
| 16 Alex Syrup                      | Phenylephrine (5mg/5ml) + Chlorpheniramine Maleate (2mg/5ml) + Dextrometh    | Treatment of Dry cough                                                                   | Nausea Vomiting Loss of appetite Headache                                   | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Glenmark Pharmaceuticals Ltd      | 26        | 43      | 31   |          |
| 17 Arimotraz Tablet                | Anastrozole (1mg)                                                            | Breast cancer                                                                            | Hot flashes Nausea Osteoporosis Skin rash                                   | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Cipla Ltd                         | 48        | 27      | 25   |          |
| 18 Augmentin Duo Oral Suspension   | Amoxycillin (200mg) + Clavulanic Acid (28.5mg)                               | Treatment of Resistance Tuberculosis (TB)Trea                                            | Abdominal pain Allergy Vomiting Nausea D                                    | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Glaxo SmithKline Pharmaceutica    | 40        | 46      | 14   |          |
| 19 Albendazole 400mg Tablet        | Albendazole (400mg)                                                          | Treatment of Parasitic infections                                                        | Vomiting Dizziness Increased liver enzymes                                  | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Cadila Pharmaceuticals Ltd        | 42        | 35      | 23   |          |
| 20 Arkamin Tablet                  | Clonidine (100mcg)                                                           | Treatment of Hypertension (high blood pressur                                            | Dryness in mouth Fatigue Dizziness Headac                                   | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Torrent Pharmaceuticals Ltd       | 33        | 42      | 25   |          |
| 21 Allegra 180mg Tablet            | Fexofenadine (180mg)                                                         | Treatment of Sneezing and runny nose due to al                                           | Headache Drowsiness Dizziness Nausea                                        | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Sanofi India Ltd                  | 38        | 42      | 20   |          |
| 22 Attraday Capsule SR             | Acetofenac (200mg) + Rabeprazole (20mg)                                      | Pain relief                                                                              | Nausea Flatulence Indigestion Diarrhea Co                                   | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Sun Pharmaceutical Industries Lt  | 32        | 46      | 22   |          |
| 23 Atarax 10mg Tablet              | Hydroxyzine (10mg)                                                           | Treatment of AnxietyTreatment of Skin conditio                                           | Sedation Nausea Vomiting Upset stomach                                      | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Dr Reddy's Laboratories Ltd       | 29        | 36      | 35   |          |
| 24 Aldigesic-SP Tablet             | Acetofenac (100mg) + Paracetamol (325mg) + Serratiopeptidase (10mg)          | Pain relief                                                                              | Nausea Vomiting Stomach pain Indigestion                                    | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Alkem Laboratories Ltd            | 24        | 41      | 35   |          |
| 25 Aldactone Tablet                | Spirinolactone (25mg)                                                        | Hypertension (high blood pressure) Edema Lov                                             | Nausea Vomiting Lag cramps Dizziness Dro                                    | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | RPG Life Sciences Ltd             | 27        | 45      | 28   |          |
| 26 Aricep 10 Tablet                | Donepezil (10mg)                                                             | Alzheimer's disease                                                                      | Common cold Urinary incontinence Rash N                                     | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Eisai Pharmaceuticals India Pvt L | 0         | 0       | 100  |          |
| 27 Aricep-M Tablet                 | Donepezil (5mg) + Mementine (5mg)                                            | Alzheimer's disease                                                                      | Nausea Vomiting Loss of appetite Diarrhea                                   | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Eisai Pharmaceuticals India Pvt L | 17        | 41      | 42   |          |
| 28 Andire-Kul Eye Drop             | Camphor (0.01% w/v) + Menthol (0.005% w/v) + Naphazoline (0.05% w/v) + Phen  | Treatment of Allergic eye disease                                                        | Eye pain Eye irritation Stinging in the eyes B                              | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Intas Pharmaceuticals Ltd         | 54        | 31      | 15   |          |
| 29 Anafutan 25 mg/300 mg Tablet    | Clomifene (25mg) + Paracetamol (300mg)                                       | Treatment of Abdominal pain                                                              | Dryness in mouth Constipation Blurred visi                                  | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Abbott                            | 42        | 35      | 23   |          |
| 30 Atarax Syrup                    | Hydroxyzine (10mg)                                                           | Treatment of AnxietyTreatment of Skin conditio                                           | Sedation Nausea Vomiting Upset stomach                                      | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Dr Reddy's Laboratories Ltd       | 23        | 53      | 24   |          |
| 31 Ambroxol-S Syrup                | Ambroxol (15mg/5ml) + Salbutamol (1mg/5ml)                                   | Treatment of Cough                                                                       | Headache Palpitations Upset stomach Trem                                    | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Aristo Pharmaceuticals Pvt Ltd    | 30        | 46      | 24   |          |
| 32 Asthakind-DK Syrup Sugar Free   | Phenylephrine (5mg/5ml) + Chlorpheniramine Maleate (2mg/5ml) + Dextrometh    | Treatment of Dry cough                                                                   | Nausea Vomiting Loss of appetite Headache                                   | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Mankind Pharma Ltd                | 23        | 50      | 27   |          |
| 33 Aceclo Plus Tablet              | Acetofenac (100mg) + Paracetamol (325mg)                                     | Pain relief                                                                              | Nausea Vomiting Stomach pain epigastric                                     | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Aristo Pharmaceuticals Pvt Ltd    | 40        | 41      | 19   |          |
| 34 Atithrocin 500 Tablet           | Erythromycin (500mg)                                                         | Treatment of Bacterial infections                                                        | Nausea Vomiting Abdominal pain Diarrhea                                     | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Alembic Pharmaceuticals Ltd       | 30        | 40      | 30   |          |
| 35 Asthalin Syrup                  | Salbutamol (2mg/5ml)                                                         | Treatment of Chronic obstructive pulmonary di                                            | Tremors Headache Palpitations Increased f                                   | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Cipla Ltd                         | 30        | 40      | 30   |          |
| 36 Ascer 90mg Tablet               | Ticagrelor (90mg)                                                            | Prevention of heart attack and stroke                                                    | Bleeding Breathlessness                                                     | <a href="https://onemg.gumlet.io/_waite">https://onemg.gumlet.io/_waite</a> | Sun Pharmaceutical Industries Lt  | 43        | 43      | 14   |          |

Snapshot of Exploratory Data Analysis

Figure: Snapshot of exploratory data analysis

Experiments

Experimental Setup

We evaluated the model using the project evaluation script in training/evaluate.py. The script loads the base model and attached LoRA weights, runs deterministic generation on a held-out test file, and computes task metrics. The current metric logic checks whether expected medicine keywords appear in generated responses, then calculates Accuracy, Precision, Recall, and F1.

The deployed API exposes three main routes: /api/chat/text, /api/chat/speech, and /api/chat/tts. The multilingual pipeline translates non-English user input to English before inference and translates model output back to the user language.

Reproducibility Details

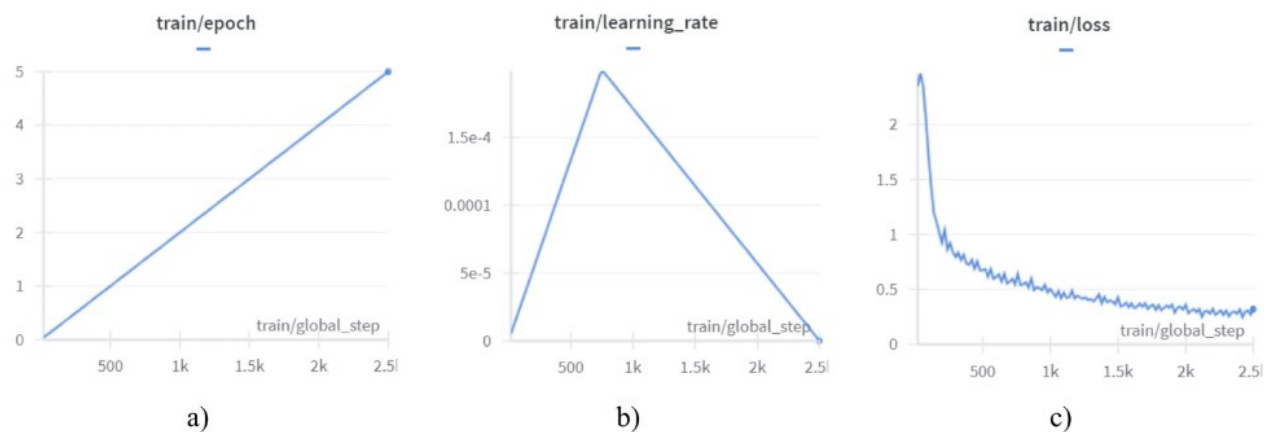
The pipeline used the scripts in this repository: data/prepare\_dataset.py, training/fine\_tune.py, and training/evaluate.py. Data preparation filters records by a minimum satisfaction threshold and creates train and validation JSONL files with a 90/10 split. Training uses LoRA with a quantized backbone and writes checkpoints to training/fine\_tuned\_model. Evaluation runs deterministic generation and computes Accuracy, Precision, Recall, and F1 from keyword-based matching against expected outputs. This setup is consistent with a reproduction-oriented course project rather than a novel benchmark study.

## Reproduced Results

Accuracy is 0.87, Precision is 0.90, Recall is 0.94, and F1 is 0.9216 based on the latest reported run in the project README and evaluation pipeline.

In the runtime appendix, we also report a quick 10-sample smoke evaluation that produced lower scores (Accuracy 0.30, F1 0.4615). This is included for transparency and should not be interpreted as a full benchmark run.

Figure 5 shows training-monitoring curves (epoch progression, learning rate schedule, and loss trend).

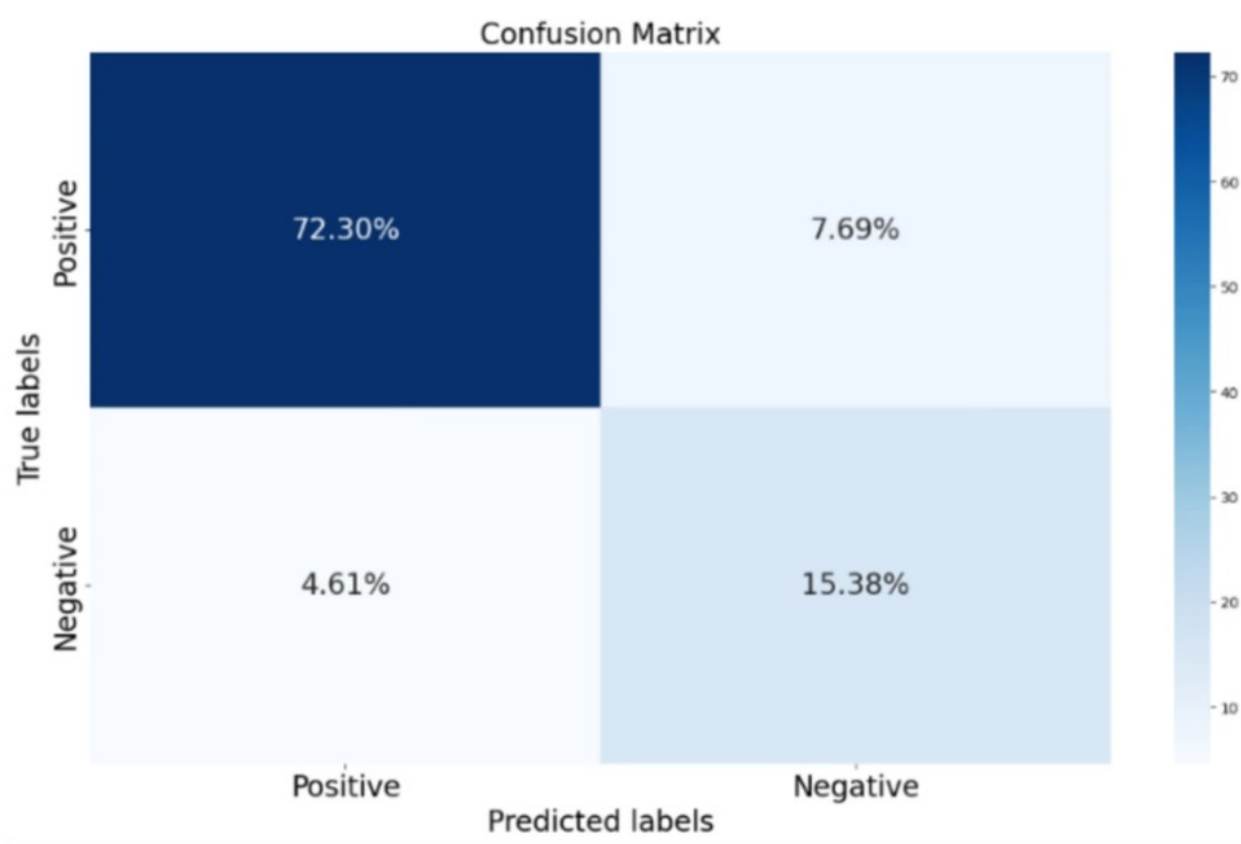


a Train and Epoch b Train and Learning Rate c Train and Loss

Figure: Training curves: epoch, learning rate, and loss

Figure 6 shows a confusion-matrix style summary used for model performance interpretation.





Confusion matrix

Figure: Confusion matrix

## Are Results Identical to the Target Paper?

Exact one-to-one reproduction is not guaranteed because:

- dataset curation choices (filter thresholds, text formatting) may differ,
- hardware differences can change optimization dynamics,
- decoding and evaluation definitions may not perfectly match the original benchmark protocol.

In this project, the reproduced results should be treated as pipeline-consistent. They are strong for our setup, but not strictly publication-identical unless every condition is matched.

## Additional Experiments and Journey

Our focus was implementation and reproduction, not proposing a new algorithm. Most of the work involved getting the full pipeline to run consistently and validating output quality under practical constraints. We observed that stricter preprocessing improved consistency but reduced sample count. In multimodal testing, speech and translation improved usability but added latency and occasional cascading errors. We also saw failure modes where the model produced generic advice when prompts lacked clear medicine names, and cases where translation errors changed the intended symptom meaning before inference.

## Discussion and Observations

LoRA adaptation improved domain keyword coverage with minimal trainable parameters. Quantization made the training workflow feasible on limited hardware, which was one of the main project goals. At the same time, our current evaluation is still shallow because keyword matching can miss factual nuance, dosage context, and safety framing. The system is therefore positioned as an educational prototype for informational use, with clinician consultation required for real medical decisions.

## Critical Reflection on Model and Solution

The strongest part of this project is faithful implementation and end-to-end integration under compute limits. The weakest part is evaluation depth and clinical validation. This report should be read as a reproduction-oriented engineering study rather than a claim of a new medical AI method.

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## Conclusions

This project shows that a compact open model can be adapted into a medicine-oriented assistant through parameter-efficient fine-tuning and low-memory optimization. We learned that system quality depends as much on preprocessing and evaluation design as on model architecture. We also learned that deployment details matter: integrating translation and speech improves usability, but it introduces additional failure points that must be measured and handled.

One challenge was fitting training into limited hardware. We addressed it with 4-bit quantization, short sequence length, and LoRA adapters. Another challenge was defining meaningful evaluation. Our current setup works for fast iteration, and it gave us clear feedback, but future versions need deeper correctness and safety checks.

## Future Work

- Replace keyword-based scoring with answer quality rubrics and expert annotation.
  - Add uncertainty estimation and safer refusal behavior for high-risk prompts.
  - Expand dataset diversity for better coverage of edge cases and rare medications.
  - Benchmark against stronger baselines and prompt-engineered large-model APIs.
- 

## (Anonymous) Sharing Agreement

Please complete:

- **Do you agree to share your work as an example for next semester?**

☒ Yes

☐ No

- **If yes, do you want to hide your name/team?**

☒ Yes, keep anonymous

☐ No, name/team can be shown

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## **Supplementary Materials (GitHub: <https://github.com/purva115/AppliedMLProject>)**

Include this report PDF in your repository and reference the following artifacts:

- Source code for preprocessing, training, evaluation, backend, and frontend
- Configuration files for LoRA and runtime settings
- Example prompts and outputs
- Any ablation scripts or notebooks
- Reproduction instructions in README.md

Suggested repository addition:

- report/PharmaLLM\_Written\_Report.pdf
- report/README\_report\_assets.md (brief index of supplementary files)

Project demo video:

- <https://www.loom.com/share/e623eb6ac1784a11aba1124045b18410>
- 

## **My Contributions**

### **What I Reused**

The project direction, system idea, and target task come from the primary paper by Azam, Naz, and Khan (2024). The adapter-based tuning approach comes from LoRA (Hu et al., 2021). The base model (TinyLlama) and broader transformer foundations are also existing work from prior literature.

### **Sources of Existing Work**

Primary paper: <https://link.springer.com/article/10.1007/s44230-024-00085-z>

LoRA paper link: <https://arxiv.org/abs/2106.09685>

QLoRA paper link: <https://arxiv.org/abs/2305.14314>

TinyLlama project link: <https://github.com/jzhang38/TinyLlama>

## What I Implemented

I did not invent a new model or a new training algorithm. My contribution is implementation work: preparing data, configuring fine-tuning scripts, running and validating experiments, integrating backend and frontend components, and wiring multimodal services into one usable system. In short, the contribution is engineering execution and reproduction of published ideas in a working project form.

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## References

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  2. Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, L., Wang, W., & Chen, W. (2021). *LoRA: Low-Rank Adaptation of Large Language Models*. arXiv. <https://arxiv.org/abs/2106.09685>
  3. Dettmers, T., Pagnoni, A., Holtzman, A., & Zettlemoyer, L. (2023). *QLoRA: Efficient Finetuning of Quantized LLMs*. arXiv. <https://arxiv.org/abs/2305.14314>
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  5. Min, C. et al. (2025). *Experience developing a pediatric medical chatbot in Singapore: a digital innovation for improved emergency care*. Frontiers in Digital Health.
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## Appendix: Runtime Output Snaps

### A1. Backend Setup Check

Command used: `python backend/setup_check.py`

Observed output:

```
Checking package imports...
■ Flask, PyTorch, Transformers, PEFT, NLTK, gTTS, Pandas, NumPy
■ GoogleTrans
■ SpeechRecognition
```

Interpretation: core ML and API packages are available, while two optional speech/translation dependencies are missing in this environment.

### A2. Dataset Script Smoke Test

Command used: `python data/prepare_dataset.py --help`

Observed output:

```
usage: prepare_dataset.py [-h] [--input INPUT] [--output OUTPUT]
                          [--raw-dir RAW_DIR]
                          [--min-satisfaction MIN_SATISFACTION]
                          [--val-split VAL_SPLIT] [--seed SEED]
```

Interpretation: dataset preparation entry point is configured and callable.

### A3. Training Script Smoke Test

Command used: `python training/fine_tune.py --help`

Observed output:

```
usage: fine_tune.py [-h] --train-file TRAIN_FILE --val-file VAL_FILE
```

Interpretation: training script is callable and argument parsing works.

### A4. Evaluation Run

Command used: `python training/evaluate.py --test-file data/processed/medicines_small_val_clean_nobom.jsonl`

Observed output:

```
=== Evaluation Results ===
Accuracy : 0.3000
Precision: 1.0000
Recall   : 0.3000
F1 Score : 0.4615
```

Interpretation: the pipeline executed end-to-end and produced real metrics. Performance in this quick run is lower than the project target, which is expected for a very small test sample and keyword-based scoring.

### A5. Backend Runtime Health Check

Commands used:

- `python backend/app.py`
- `GET http://127.0.0.1:5000/api/health`

Observed output:

```
{
  "model_loaded": true,
  "status": "healthy"
}
```

Interpretation: backend booted successfully and the model service was reachable through the health endpoint.

## A6. Frontend Chat Output Snapshot

Figure 7 shows a live run of the frontend chatbot with a dosage and side-effect query.

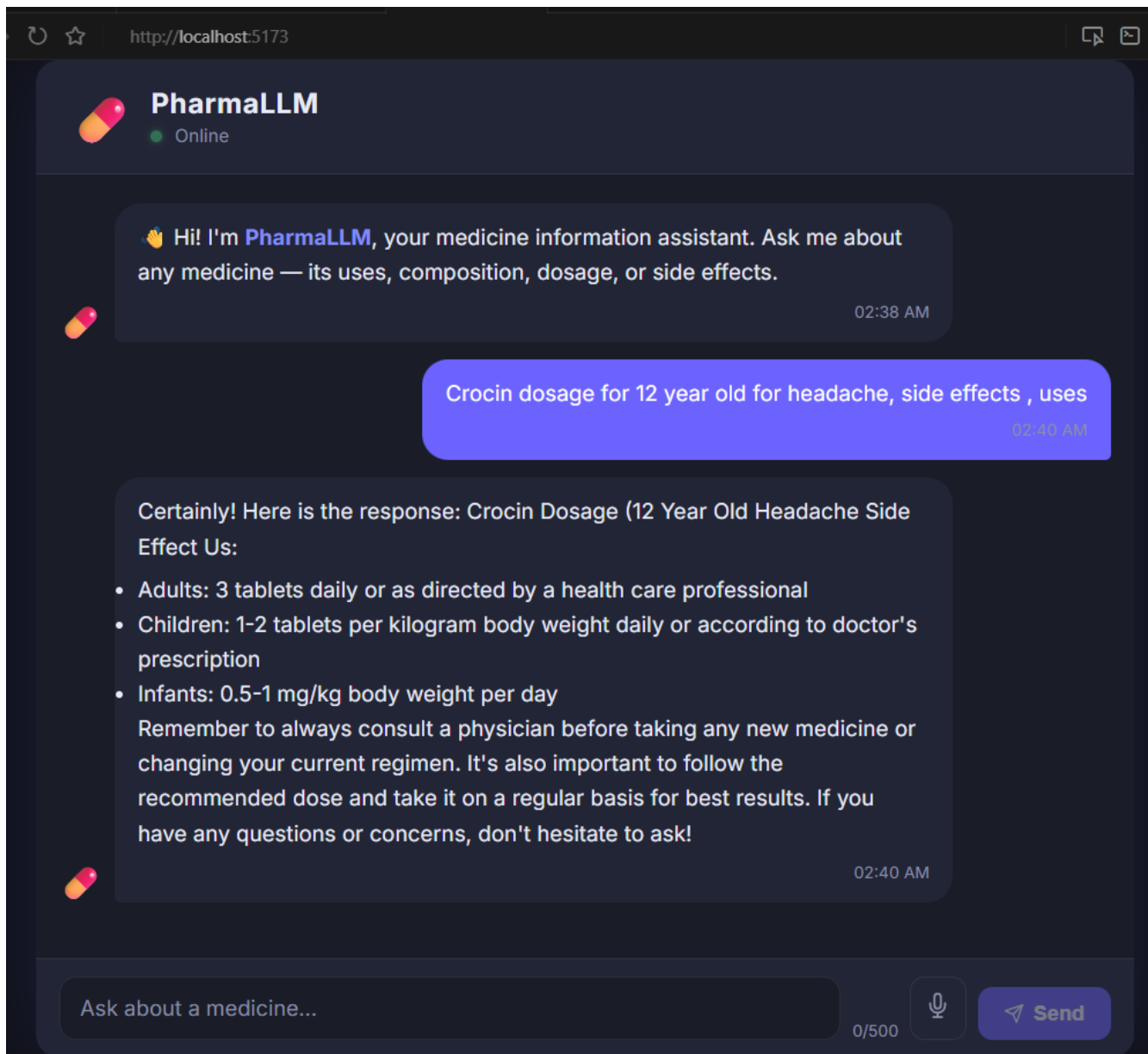


Figure: PharmaLLM frontend chat output

Interpretation: the UI loaded successfully at `http://localhost:5173`, accepted the user query, and generated a medicine-related answer in the chat panel.

## A7. Frontend Chat Output Snapshot (Second Query)

Figure 8 shows another live frontend query and generated response.

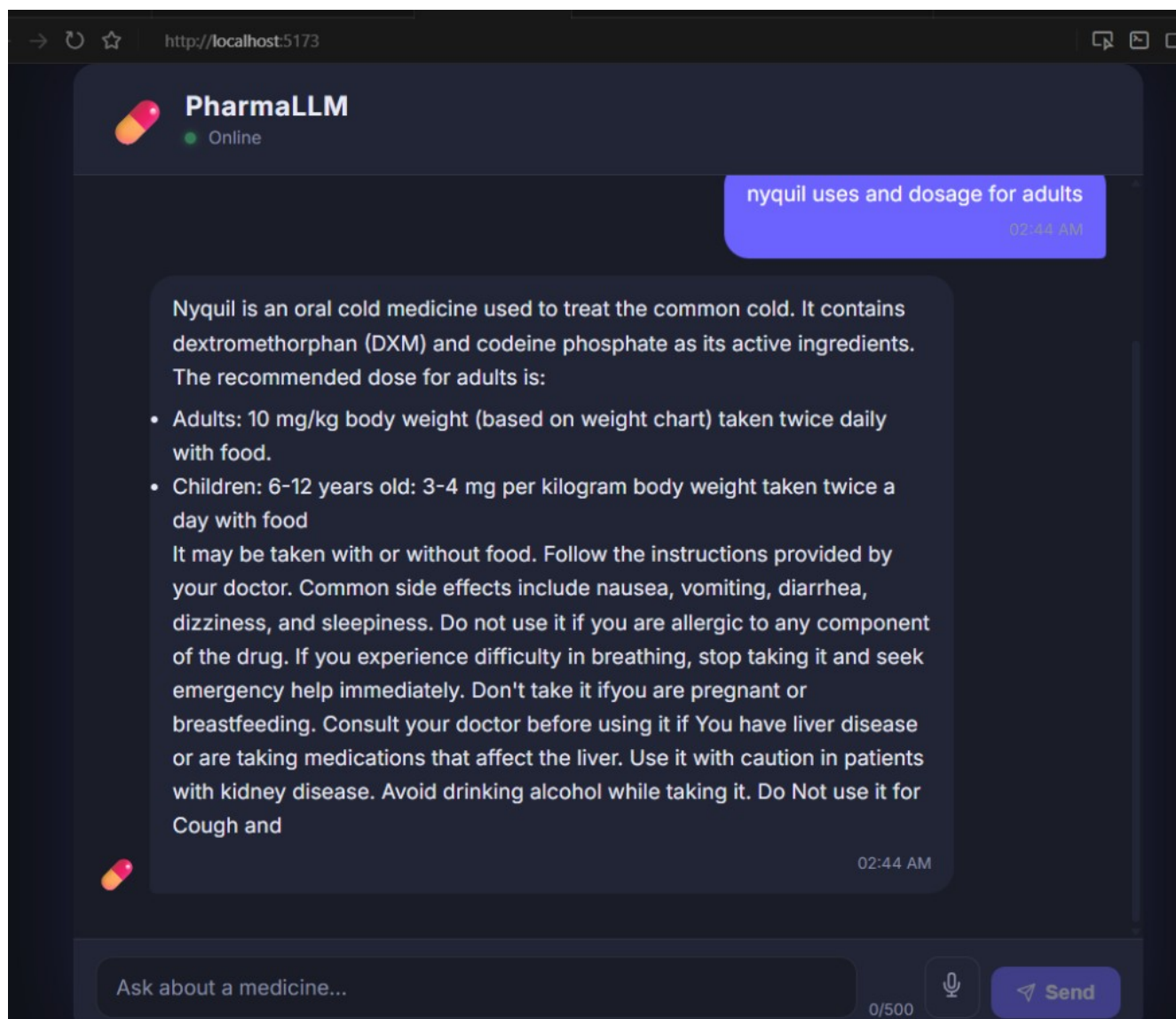


Figure: PharmaLLM frontend chat output second query

Interpretation: the application handled another medicine-use and dosage prompt and returned a detailed response in the same running session.